



Interpreting and visualizing the unit of measurement in the Rasch Model

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ABSTRACT

One of the more attractive features of the Rasch Model as a paradigm for measurement in the human sciences is that it has the potential to locate respondents on a test or survey instrument on a scale with equal intervals. Wright Maps (also known as person-item maps) are sometimes used as graphical visualizations of such scales, but they are frequently very difficult to interpret. One reason for this is that they are premised on an arbitrary unit of measurement, the logit. In this article an approach is illustrated whereby a new reference unit can be established for the interpretation and visualization of differences along the scale. This new unit is premised on the substantive choice of two locations on the scale that represent benchmark differences in the psychological attribute of interest. The invariance of the unit to differences in the context in which the measuring instrument was administered is then subject to empirical evaluation.

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The graduated ruler is a canonical visual representation of a measurement procedure applicable to a physical attribute. One reason for this is the tangible sense a ruler provides for the different units with which it is possible to realize measurement outcomes. The same ruler will often display units graded in inches along one side, with millimeters and centimeters represented on the other side. The notion of a ruler is sometimes used as an analogy for the kind of measurement that would be desirable in the context of psychological attributes. Along these lines it has been argued that when the ordered responses of subjects to survey or test items can be fit by one of the mathematical functions associated with the Rasch family of item response models [25,35,2], it becomes meaningful to visualize the results graphically using units—the logit—that mimic the role of an inch or centimeter in the graduated ruler (e.g., [5,23]).

Graphic visualizations of Rasch Model parameter estimates have been given the name “person item maps” by some (e.g., David Andrich and developers of the RUMM software) and Wright Maps¹ by others [34]. Fig. 1 is a Wright Map depiction taken from the book *Rasch Analysis in the Human Sciences* by Boone, Staver & Yale [6]. This instance of a Wright Map is being used to visualize the results after fitting a Rasch Model to the data that arise from 75 science teachers responding to a survey in which they have been asked to agree or

disagree with 13 statements related to their feelings of self-efficacy. The numbers that run from −2 to 6 up the lefthand column represent the units of the logit scale. The symbol “X” denotes the estimated location of a unique respondent on the scale; those located higher/lower on the continuum are those with greater/lesser amounts of self-efficacy. The rightmost column comprised of alphanumeric characters depicts the survey items to which the teachers responded, located on the same logit scale. The items that are higher/lower on the continuum are those statements related to self-efficacy that were harder/easier for respondents to endorse. The vertical dashed line separates the person side from the item side of the visualization. The letters M, S and T characterize, respectively, the mean, and one and two standard deviations away from the mean for two different distributions: person locations and item locations. This Wright Map is on the whole fairly typical of the results that often accompany applications of the Rasch Model, whether these are presented as part of a textbook introducing the Rasch Model, in journal articles, reports, or conference presentations.

As an analogy to measurement with a ruler, or perhaps to measurement using a thermometer, Wright Map visualizations such as these have potential, but leave considerable room for improvement. Tufte (1989) suggests that “graphical excellence” in the context of the presentation of quantitative information can be understood as (1) well-designed presentation of interesting data, (2) complex ideas communicated with clarity, precision and efficiency, and (3) that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space. Many, if not most Wright Maps fall short on all three of these criteria. One reason is that when these sorts of graphics are

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¹ I will use the term Wright Map throughout both for the simple reason that it requires two words instead of three, and because it seems fitting that I pay homage to my former advisor Mark Wilson by using the term he invented to pay homage to his former advisor Ben Wright.

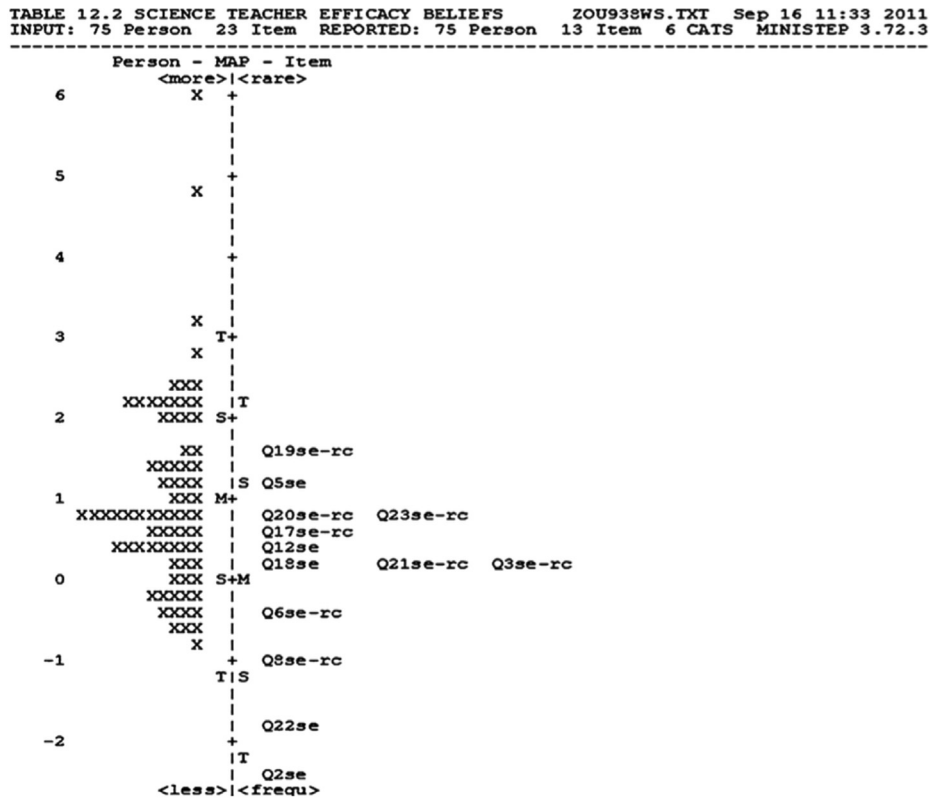


Fig. 7.1 A Wright Map of 75 respondents who responded to the 13 self-efficacy items of the STEBI. Persons are plotted on the *left side* of the vertical line, and items are plotted on the *right side* of the vertical line. In this analysis, persons with a higher measure are those persons who were more agreeable to survey items. Items with a higher measure are items that were harder for respondents to agree with

Fig. 1. A Common Depiction of a Wright Map, from Boone, Staver & Yale [6].

constructed, not enough thought has been given to the purpose of the graphic, the intended target audience, and the setting in which the graphic will be viewed. Graphic design requires attention to aesthetics, and the essential aesthetic features of effective Wright Maps (and for that matter, any graphic representation of measurements taken of non-physical attributes), is an issue that has been largely overlooked. However, beyond questions of aesthetics, the interpretability of many Wright Maps for anyone other than their creators might be symptomatic of a deeper issue: a lack of clarity and intentionality in the definition of the scale and the scale's unit of measurement. In this paper a novel approach is suggested that might help address the problem both in the context of graphical visualizations of Rasch Model results, and in the context of using the model to interpret the answers to questions about differences in the magnitudes of psychological attributes. To keep things internally consistent, it is assumed throughout that the Rasch Model is being applied in a context for which it is plausible to hypothesize an underlying quantitative attribute. Given this hypothesis, the focus of this paper pertains most directly to what Michell [20–25] would describe as the “instrumental task” of measurement, but as Sherry [27] points out, the scientific and instrumental tasks of measurement are typically interwoven.

1. The unit of measurement in psychometric models

Humphry [14–18] has written extensively on the different roles of the unit in physics and psychometrics, and though I will not

revisit all the details of these expositions, I adopt a similar notational framework. For convenience, imagine a scenario in which students take a test comprised of unique items. Let X_{pi} represent a discrete random variable with two possible values, $X_{pi} = 1$ if student p answers item i correctly, otherwise $X_{pi} = 0$. The probability of a correct response can be written according to a logistic function as

$$P(X_{pi} = 1) = \frac{\exp(\rho_s(\theta_p - \beta_i))}{1 + \exp(\rho_s(\theta_p - \beta_i))}, \quad (1)$$

where β_i is scale location (i.e., difficulty) associated with item i , and θ_p is the scale location (i.e., ability) of student p . The term ρ_s represents what Humphry describes as a general discrimination parameter that is “common across classifications or levels $s = 1, \dots, S$ of an empirical factor.” By empirical factor, Humphry means some common defining feature of a subset of items that allow for the interpretation of distances along an underlying measurement scale with respect to a common unit. The expression can be simplified when reformulated in terms of logits, as in

$$\text{logit}(X_{pi}) = \ln \left(\frac{P(X_{pi} = 1)}{P(X_{pi} = 0)} \right) = \rho_s(\theta_p - \beta_i). \quad (2)$$

With this formulation, it is evident that distances in the locations of two different students on the scale, or two different items on the scale, depend on the magnitude of ρ_s . That is, consider the difference in location of two students relative to any item

$$\text{logit}(X_{1i}) - \text{logit}(X_{2i}) = \rho_s(\theta_1 - \theta_2). \quad (3)$$

Likewise, the difference in location of two items relative to any student is

$$\text{logit}(X_{1i}) - \text{logit}(X_{2i}) = \rho_s(\beta_1 - \beta_2). \quad (4)$$

The parameter ρ_s defines the unit of measurement of the scale, where larger values are indicative of an instrument with finer discrimination. Consider two possible cases, one in which $s = i$, and the other where $s = 1$. When $s = i$, the general logistic measurement function becomes an instance of the two-parameter logistic model (2PLM; [19,4]). A well-known problem with the 2PLM as a model for measurement is that the unit of the scale is not invariant to choice of items on the instrument. Indeed, if we conceptualize the items on a test as eliciting repeated measures of a single underlying psychological attribute, then each item in a 2PLM can, in principle, enact this with a unique scale of measurement. In contrast, when $s = 1$, the general logistic measurement function becomes an instance of the Rasch Model. A defining feature of this model, illustrated in Eqs. (3) and (4) above, is what Rasch referred to as specific objectivity [25,26]. Provided that the model fits the data, the difference in the location of two any students on the scale is invariant to the particular set or subset of items that were administered to them. Symmetrically, the difference in the scale location of any two items is invariant to the particular set of students that responded to them. The challenge of applying the Rasch Model is one of design, as it is no easy task to create items according to a single well-established empirical factor such that they will only differ by intention with respect to their difficulty while maintaining a constant degree of discrimination.

The role of the unit of measurement is often obscured because it has become conventional both in written presentation and in computational estimation to constrain $\rho_s = 1$ such that

$$\ln \left(\frac{P(X_{pi} = 1)}{P(X_{pi} = 0)} \right) = \theta_p - \beta_i. \quad (5)$$

When imposed, this constraint has the effect of shifting information about discrimination to the variability of the θ_p and β_i distributions, $\sigma^2(\theta)$ and $\sigma^2(\beta)$. It is for this reason that comparisons among groups of students and items are typically expressed in “effect size” units by dividing a mean difference in logits by $\sigma(\theta)$ or $\sigma(\beta)$.

The interplay between the unit of measurement ρ_s and the standard deviation of the θ_p and β_i distributions can be a recipe for confusion when two different samples of students are being compared on two different measurement instruments unless it is known in advance that each instrument has the same constant value of ρ_s , or that each sample has been drawn at random from the same population. If, instead, the two samples come from populations that could differ in their variability, and are administered tests that could vary their discriminability, observed differences in variability across the two tests can be conflated with differences in the units of measurement. Calibrating the Rasch Model for two different tests taken by two different samples under the constraint that $\rho_s = 1$ only appears to equate the units of measurement, just as calibrating the two different samples under the constraint that $\sigma^2(\theta) = 1$ only appears to equate the variability.

All of this represents an issue that is well-understood in the psychometric literature, and a variety of designs and adjustments have been proposed and enacted in the attempt to make the results from different tests or different populations comparable

on a common scale [9,17]. I would argue that what remains a deeper problem, even when these attempts are successful, is the lack of interpretability of the conventional unit of measurement in the Rasch Model, the logit. Humphry reminds us that the in metrology a measurement unit is a “scalar quantity, defined and adopted by convention, with which any other quantity of the same kind can be compared to express the ratio of the two quantities as a number” [39]. For example, the pure number $2.734 = L: m$ is the ratio of the length $L = 2.734$ to the length $1 m$. [14].

In furthering the analogy to measurement under the Rasch Model, Humphry expresses the discrimination parameter ρ_s as a ratio of two units, $[\mu_+]$ and $[\mu_s]$, where $[\mu_+]$ is intended to represent the “common unit” for the underlying attribute to be measured, and $[\mu_s]$ is the unit specific to a subset of items chosen for the measuring instrument. Hence, when ρ_s is fixed to 1, it can be taken as a claim that all items in the subset measure in terms of the same common unit. Unfortunately, unlike the meter, a logit is an abstract mathematical entity, one that at best establishes the unit of an interval scale [11]. To state the obvious, one does not interpret the measure of a person in n logits (where n is a real number) as the ratio of the amount of the attribute to a single logit. Rather, the logit is only meaningful in its potential to facilitate comparisons of *differences* between people in terms of distances between the locations of items (or between items in terms of the distances between the locations of people). The implication of this for Wright Map visualizations of a Rasch Model is that there is no a priori reason to expect 1 logit to represent a substantively meaningful difference between people or items, and this makes differences in logit units difficult to interpret. The problem here is somewhat analogous to measuring temperature in celsius degrees without having established the physical meaning of 0 and 100 as substantive benchmarks. In what follows I describe an approach that could be taken to establish these benchmarks in an educational context, and thereby define a targeted reference unit for differences that is some designated ratio of a logit.

2. Illustrative data context

2.1. The Genetics Concept Assessment

The data at hand come from a pre-post administration of the Genetics Concept Assessment (GCA; [29]) to a total of 1667 undergraduate students enrolled in a semester-long introductory genetics courses offered at six different American post-secondary institutions. The GCA is a test comprised of 25 multiple-choice items wherein the incorrect response options are intentionally chosen to represent common student misconceptions. The items were designed to map to 8 learning goals common among introductory courses in genetics and have also been coded for cognitive complexity according to the levels of Bloom's Taxonomy [18]. The GCA was administered as a pre-post assessment in unique semester-long courses taught between the Fall of 2015 and the Spring of 2017.

Fig. 2 summarizes the overall results when each student's GCA score is computed as the count of GCA items answered correctly. On average, students answered roughly 10 items correctly on the pre-test and 17 on the post-test for a gain of 7 items answered correctly. The classical reliability of scores on each occasion, estimated using the alpha coefficient, is 0.63 (pre) and 0.78 (post). Notably, but not unexpectedly, the reliability of student-level gain scores, 0.56, is less than the reliability for level scores on either occasion.

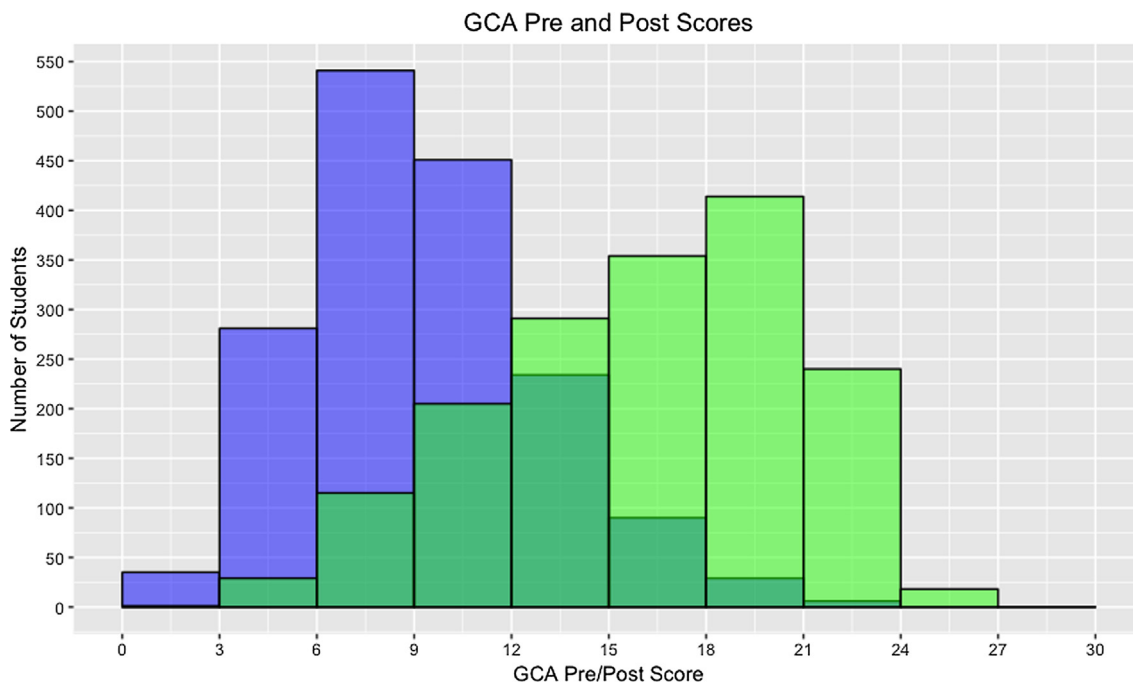


Fig. 2. Histograms of GCA Score Distributions. Blue Shading represents pre-test scores, green shading post-test scores. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Fitting the Rasch Model and creating visualizations

Fitting the GCA data to the Rasch Model will not change the high-level interpretation of these results, which are indicative of student learning. If the items of the GCA represent valid learning objectives, then the shift in the distribution from pre to post appears significant. On the other hand, the magnitude of any test score gain expressed as the number or percent of items answered correct is somewhat arbitrary, since this depends on both the ability of the students and the difficulty of the items. Strategies often taken to express the raw gains in standard deviation units or in percent of total points possible do not address this fundamental issue. An argument—perhaps the best argument—for applying the Rasch Model in this kind of a research context is that provided that the model fits the data, it allows for the construction of a scale for measurement with an invariant reference unit.

GCA item responses were fit to the Rasch Model using the package `TAM` [37] within the R computing environment [24]². Item parameters for pre and post GCA responses were estimated with the constraint that the origin of the logit scale corresponds to the mean of the item difficulty distribution. Person ability parameters are then estimated in a second step using weighted likelihood estimation [32]. Although judgments about item (and person) fit to the Rasch Model are, unfortunately, never straightforward [36] or conclusive [22], all but two items showed good fit, and for the remainder of the paper, I proceed under the assumption that the fit of the model is adequate for illustrative purposes.

² The `TAM` package contains functionality that overlaps considerably with the proprietary ConQuest software package [1], but has the advantage of being free and open source. Other R packages such as `mirt`, `ltm`, and `eRm` could also have been used to fit the GCA data to the Rasch Model. The packages `TAM`, `ltm` and `mirt` all default to marginal maximum likelihood estimation of item parameters; the package `eRm` uses conditional maximum likelihood estimation. For the illustrative context in this paper, the choice of package and estimation algorithm has no significant impact on item or person parameter estimates and can be ignored as a factor of interest.

3. Establishing scale benchmarks for reference units

Fig. 3 depicts a conventional Wright Map display of the results from calibrating the GCA post-test with the Rasch Model³. The left-hand side of the map presents the distribution of student locations, while the righthand side of the map presents the distribution of item locations. The student distribution is expressed as a histogram with class intervals of 0.5 logits, the item distribution is a histogram with specific labels for each item and class intervals of 0.25 logits. As the scale has been anchored by constraining the mean of the item locations to be 0, the Wright Map conveys that the average student in the sample, with a location at 0.87 logits, has a 70% chance of answering an average GCA item correctly. The asymmetry in the two distributions also conveys that the GCA is more targeted to students at the lower half of the ability distribution than the upper half. What this version of the Wright Map does not convey is any substantive benchmark for comparing the locations of any two students, or any two groups of students. This requires a theoretical rationale, and I consider two strategies that could be taken.

³ The visualizations in this paper are all created in R using either the `WrightMap` (Torres Iribarra & Freund, 2016) and `ggplot2` packages the base R graphics package. There are three prominent standalone software packages written specifically to calibrate data to the Rasch family of item response models: `RUMM` [3], `WinSteps` [38] and `Conquest` [1]. All three can generate Wright Maps as graphical output, and each vary with respect to the ability of the user to tailor this output. An advantage of R, and the reason it is employed here, is that R graphics packages are all free and open source, and offer the greatest flexibility to the user in creating carefully tailored graphic visualizations. The `ggplot2` package is especially remarkable in this regard, and as developed by Hadley Wickham has helped to reconceptualize the activity of graphic design of data visualizations in terms of a “grammar of graphics” premised on layers [33]. Note that the flexibility of packages such as `ggplot2` and `WrightMap` is such that they can easily be used in combination with standalone packages such as `RUMM`, `WinSteps` and `ConQuest`. The script for all graphics produced in this paper are available upon request.

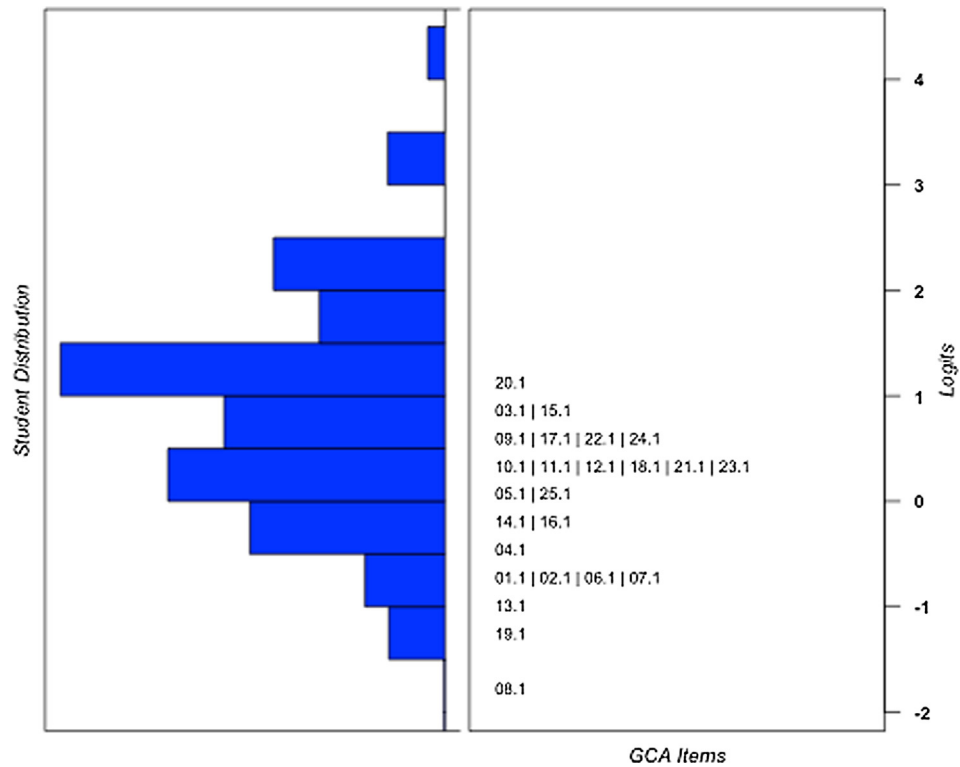


Fig. 3. A Conventional Wright Map of the for the Genetics Concept Assessment with Default Reference Units.

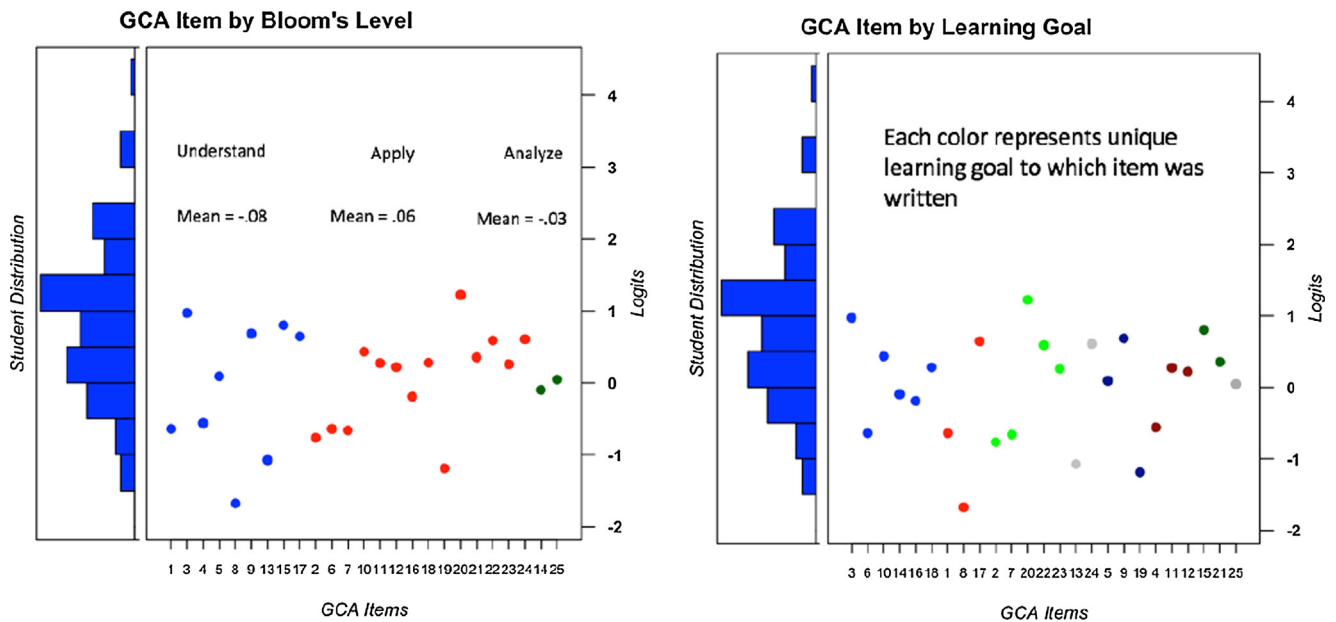


Fig. 4. Wright Maps that Focus Attention on Item Design Variables.

3.1. Strategy 1: focus on item-specific design factors

A core idea when designing items in accord with the principles of the Rasch Model is that item difficulty should be predictable by design [34]⁴. In a best-case scenario, it should be possible to predict

⁴ In the case of polytomous items, it is the location of item category intersection curves that should be predictable.

item difficulty through the use of item design variables [13,7]. This was the premise behind the Linear Logistic Test Model [10], and has also been central in applications of the Rasch Model to create a scale for reading comprehension [30]. In the present context, two item design factors could be considered as plausible predictors of GCA item difficulty: Bloom's taxonomy level and learning objective. The two panels of Fig. 4 present new Wright Map visualizations with item locations grouped by Bloom's Level (left panel) and Learning

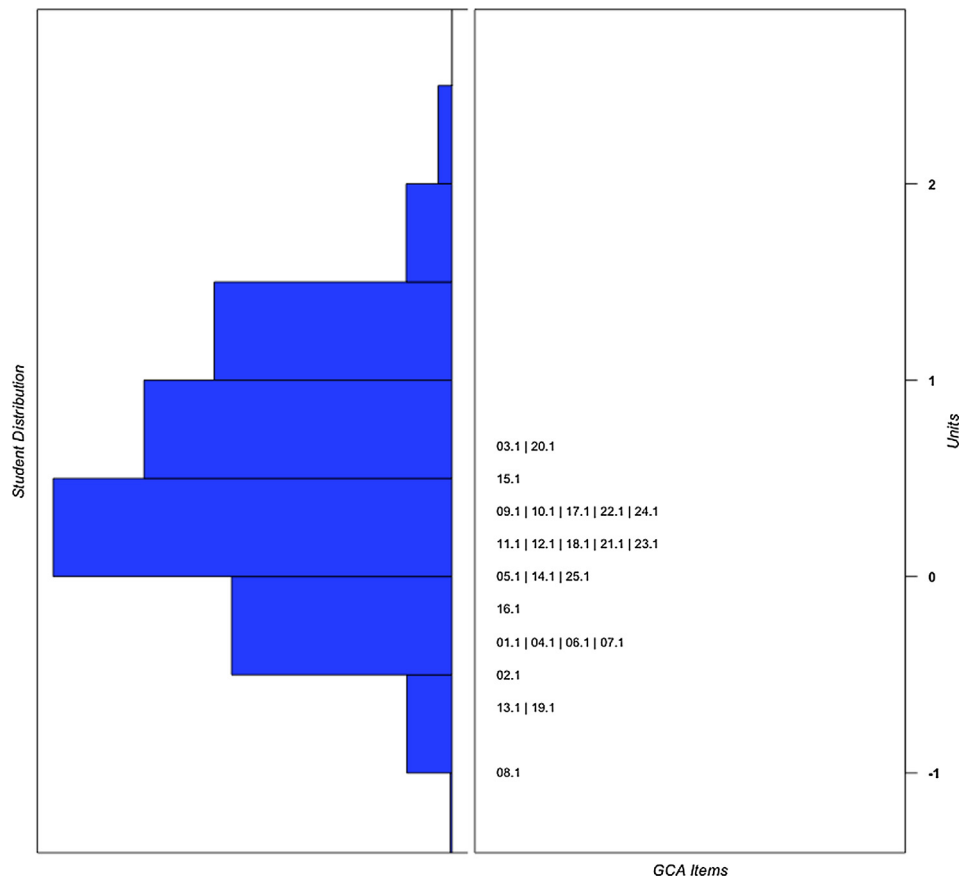


Fig. 5. Rescaled Wright Map for the Genetics Concept Assessment with Reference Units Defined as the Distance between Two Benchmark Items (24 and 13).

Goal (right panel). It is immediately apparent that neither of these factors has an association with item difficulty, and therefore neither design factor provides a good basis for item-side benchmarks⁵.

If there had been a strong association, Wright Maps such as these would invite the viewer to compare differences in the locations of students relative to differences in mean item difficulty associated with different levels of a design variable. For example, if the mean difference in locations between items written to correspond to Bloom's Levels of Apply and Understand had been, say, 0.5 logits, this would become a meaningful frame of reference when comparing distances between student locations.

3.2. Strategy 2: focus on learning targets implied by different item families

In consultation with the one of the developers of the GCA who teaches a course in introductory genetics, two items were identified that could be viewed as benchmarks for the growth in student understanding over the span of a semester. The two items, 13 and 24 (See Appendix Figure A-1), had been written to the same learning objective ("Describe the phenomenon of linkage and how it affects assortment of alleles during meiosis"), but item 13 was viewed as representing a concept that should be fairly easy for students to master (coded at Bloom's Level 2), while item 24 was viewed as far more challenging (coded at Bloom's Level 3). As it

turned out, after fitting the GCA post-test to the Rasch Model there was in fact a large difference between the two item locations, $\hat{\delta}_{13} = -1.07$, and $\hat{\delta}_{24} = .61$. From an instructional point of view, the distance in the locations of item 13 and 24 of 1.68 logits represents a magnitude that might be plausibly hypothesized to represent the typical amount of instruction it would take to move a student from being able to answer just item 13 correctly, to being able to answer both item 13 and 24 correctly.

3.3. Rescaling the unit depicted in a Wright Map

With his information in hand, we could choose to redefine the logit scale so that a new unit, what I will refer to in this context as a g-unit (g for growth), is interpretable as the distance in the locations of items 13 and 24. One way this could be done would be to calibrate the Rasch Model under the constraint that the difference between these two items equal one. Another way this could be done after the Rasch Model has already been calibrated (under the default item difficulty constraint) would be to impose a post-hoc transformation. For example, we could solve the equation $\rho_s(\delta_{24} - \delta_{13}) = 1$ given the known values of our two benchmark items and find that $\rho_s = .595$. Alternatively, one could define a reference g-unit as any desired ratio of the distance between the two items, e.g., $\rho_s(\delta_{24} - \delta_{13}) = 2$, in which case $\rho_s = 1.190$. After multiplying both student and item parameter estimates by these constants, we can generate the transformed Wright Maps shown in Figs. 5 and 6, now interpretable with respect to a g-unit.

⁵ This did not come as a surprise to the GCA developers, who did not expect certain learning objectives to be more difficult than others, and who had coded the items by Bloom's Taxonomy post hoc as opposed to designing them a priori to vary in complexity.

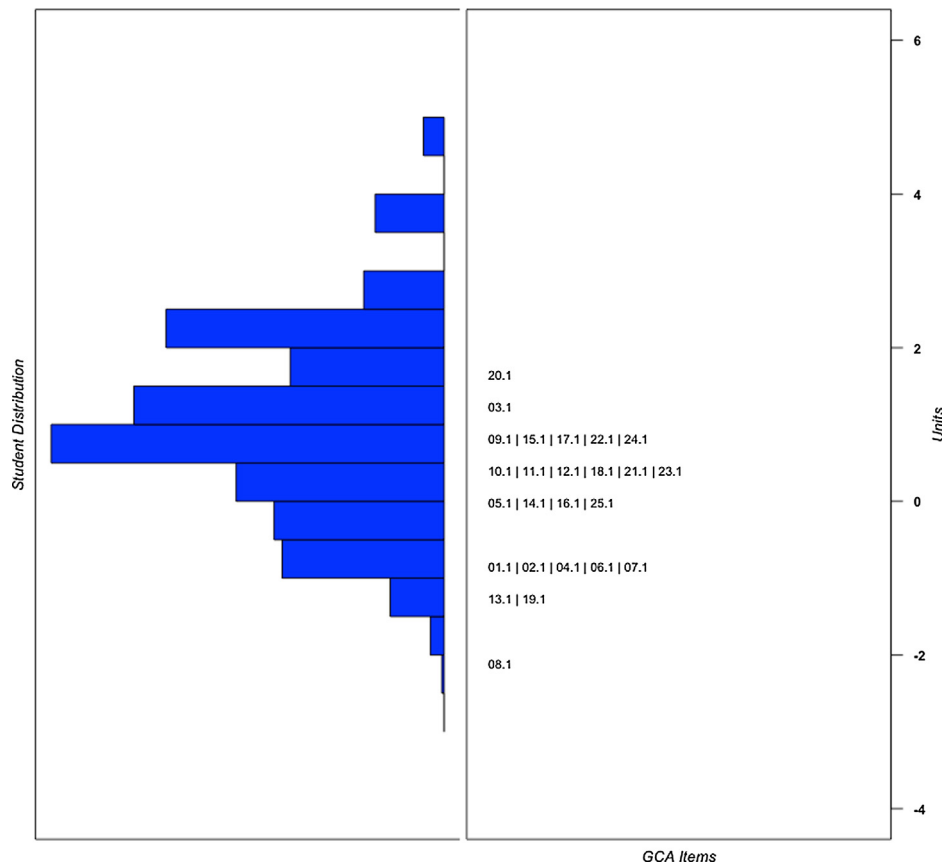


Fig. 6. Rescaled Wright Map for the Genetics Concept Assessment with Reference Units Defined as Half the Distance between Two Benchmark Items (24 and 13).

For the map in Fig. 5, the g-unit is the full difference between the locations of items 13 and 24; for the map in Fig. 6, the g-unit is half the difference. The choice of g-unit has a noticeable impact on the observed variance of the student and item distributions. As a point of comparison, under the default calibration of the Rasch Model (see Fig. 3), in which the unit of discrimination is constrained to 1 (i.e., $\rho_s = 1$) the SD of the student distribution was 1.03 and the SD of the item distribution was 0.73. When the g-unit is defined such that a new logit represents a difference of 1.68 on the old logit scale, these respective SDs decrease to 0.62 and 0.44. When the g-unit is defined such that new logit represents a difference of 0.84 on the old logit scale, the person and item SDs increase to 1.24 and 0.87. Of course, nothing about the “true” discrimination of the measuring instrument has changed. What has changed is the substantive rationale for interpreting a difference in locations. When the scale is redefined such that a reference unit for growth on the original scale is 1.68 logits, one is communicating a threshold against which observed differences between two students or groups of students should be evaluated. With this new threshold established, there will be fewer substantively meaningful differences between any two locations relative to the original scale. In contrast, when the scale is redefined such that a reference unit for growth is 0.84 logits, there will be somewhat more meaningful differences between any two locations relative to the original scale.

There are naturally many other methods that could be used to establish the two locations used to define a reference unit for growth, but the key property is that the locations are defined with respect to known cognitive features of the items with which students interact in the test experience. The reference unit is chosen

so that it will convey something substantive about a target difference in what students know and can do. Even when a reference unit is not used to transform the scale when presented in a visualization, it can be used as an implicit conversion factor. This would be akin to someone being shown a difference in distance in inches, but translating the difference into a more meaningful unit, centimeters.

3.4. Testing for invariance

That the distance between item locations should remain invariant irrespective of the sample of persons is fundamental to the Rasch Model property of specific objectivity, and it is also something that can and should be evaluated empirically. This was done for the GCA data with two genetics courses from different institutions. Both courses had roughly comparable numbers of students enrolled. The GCA items were calibrated separately for each course. Next, the item parameters for one course were placed onto the same scale as those of the other, and then the differences in the scale locations of items 13 and 24 were compared. For one course, the difference in the two items was about 2 logits, relatively close to the difference found using the full data set that included students from the courses offered by all six participating institutions. But for the second course, the difference was much smaller, just 0.90 logits. This raises some questions about the invariance of the scale intervals. For students taking courses in genetics that vary in their curricular emphases, the distances between items may take on different meaning with respect to their implications for magnitude. There is no easy solution to this challenge, which is more along the lines of what Michell [20,21] characterizes as the

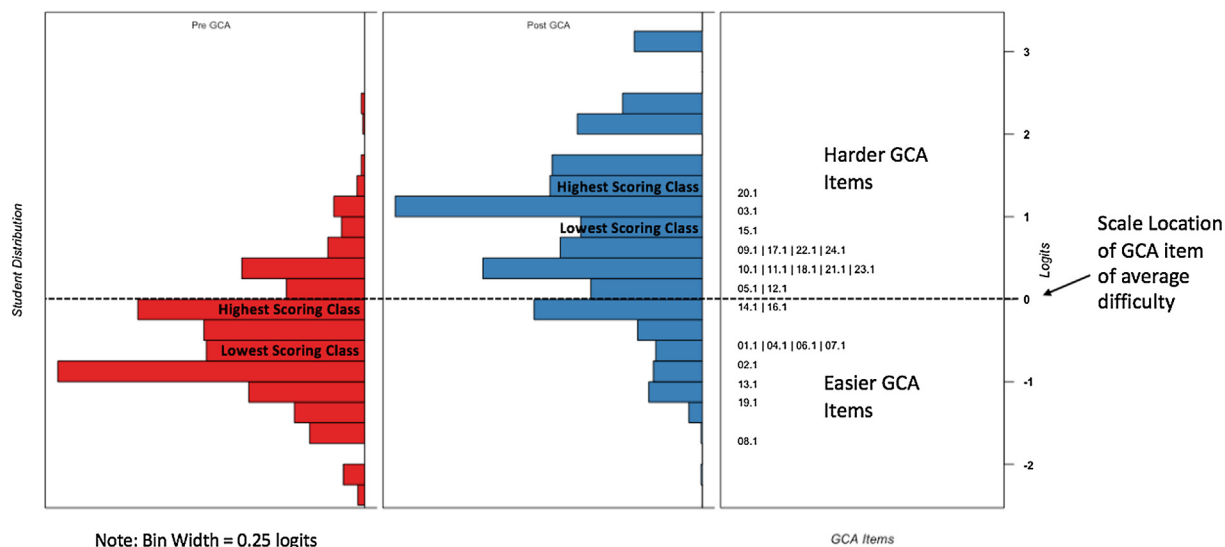


Fig. 7. Growth in GCA Ability Distributions Relative to Item Locations.

“scientific task” of measurement. The focus on defining reference units for growth with respect to an assumed instructional sequence brings this challenge from the background to the forefront.

4. New interpretations and visualizations of growth magnitude

The impetus for Georg Rasch to develop the model that bears his name was the need to place a sequence of tests given to students at different grade levels onto a common scale so that it would be possible to make comparisons about gains across grades. Establishing a reference unit for growth can direct viewers to substantive interpretations of growth magnitudes, and this can help inform judgments about the practical significance of these magnitudes. Fig. 7 is an example of a Wright Map used to convey information about growth from pre to post by presenting histograms of the GCA ability distributions on each occasion and juxtaposing the shift in these distributions against GCA item difficulty, which has been fixed here to the calibrated values from the post-test administration⁶. In this visualization, one obvious purpose is to communicate the significant shift in the student distribution. The dashed line and annotation at the origin of the scale direct the viewer to an important frame of reference, the average difficulty of a GCA item. One can see that there are really two groups of GCA items, 8 that are well below average in difficulty, 5 that are right about average, and another 12 that are well above average. Prior to instruction, about half the students had at least even odds of answering the 8 easier items correctly, but the odds of answering the harder items correctly was low for almost all students. After instruction, almost all students had at least even odds of answering items in the easier group correctly, and most also had even odds of answering the items in the harder group correctly. An additional annotation, placed within the bins of the two histograms, conveys the location of the mean logit value for the lowest and highest scoring classes on each occasion. From this we can see that these two

Table 1
Different Ways to Characterize Magnitude of Pre to Post Growth.

Metrics	Pre to Post GCA Gain	Interpretation
Raw Gain	6.8	On average students answered about 7 more GCA items correctly from pre to post.
Raw Gain in Percent	27%	On average the percent of GCA items students answered correctly when up by 27 from pre to post.
Normalized Raw Gain	44%	On average students increased GCA raw score by 44% of the maximum score they could earn
Logits	1.38	On average student understanding of genetics increased by 1.38 logits
Effect Size (Logits in SD Units)	1.50	On average student understanding of genetics increased by an amount that is 1.5 times the SD of the post GCA distribution
Logits in Reference Units for Growth (g-units)	0.90	On average, student understanding increased by about 90% of the difference in difficulty between target learning objectives in genetics

Note: Normalized Raw Gain = $100 \times (\text{Post-Pre}) / (100 - \text{Pre})$; Logit in SD Units = $(\text{Post Mean in Logits} - \text{Pre Mean in Logits}) / \text{SD of Post Distribution}$.

locations both shifted upward by about a full logit unit, but the distance between the lowest and highest scoring class on each occasion stayed about the same.

The Wright Map in Fig. 7 invites the question: how much did the average student enrolled in an introductory genetics course grow in their understanding of core concepts? Table 1 summarizes the different approaches that could be used to provide a sense for the magnitude of average growth. The first three metrics are all based on computations and transformations using the raw item scores⁷. The last three are all based on computations and transformation using logits after applying the Rasch Model. It seems worth pointing out that gains expressed in logits, taken in isolation, are probably the least interpretable of the group. In contrast, when these

⁶ There are several ways the person parameters for this sort of visualization could be estimated. One could estimate θ values for both pre and post administrations using the pre-test item parameter values each time, or using the post-test item parameters each time (the approach taken here). It would also be possible to do a joint or concurrent calibration, or a multidimensional specification. These technical issues are outside the scope of the present article, but see [11,12] and Embretson [8].

⁷ The normalized raw gain is a popular metric among those involved in disciplinary based higher education research. It is intended to adjust for the fact that students with higher pre-test scores have less “room” to show growth.

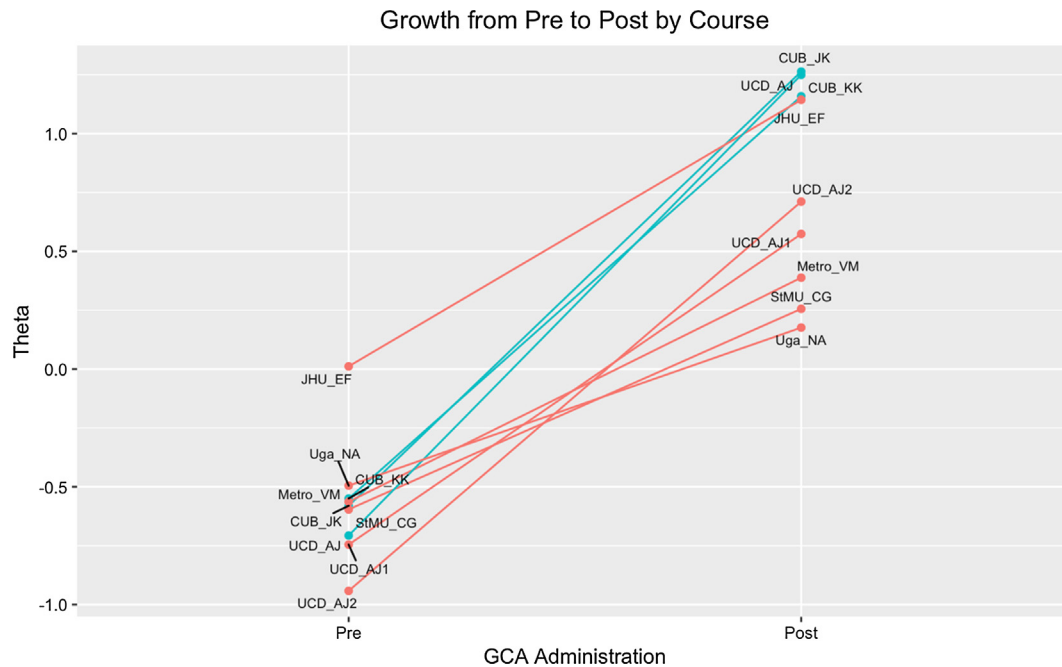


Fig. 8. Growth Visualization with Respect to Reference Unit. Red lines represent courses where gains were less than a reference unit; blue lines represent courses with gains equal to or greater than a reference unit. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

gains are expressed relative to the g-unit benchmarks introduced in the previous section, it represents the only number that conveys a growth magnitude that can be interpreted relative to the instruction students receive in genetics.

The Wright Map in Fig. 7 does not actually depict student-specific GCA gains. This could be accomplished by computing the difference in θ estimates for each student from pre to post and replacing the two histograms with a single histogram of these difference values. One reason to be careful with such representations is the fact that student-level differences will be less reliable (often much less reliable) than student locations at one point in time. An alternative is to compare differences with respect to groups of students enrolled in different genetics courses. Fig. 8 shows a visualization created using `ggplot` and inspired by Tufte [31]. This visualization directly evaluates growth magnitudes from the original scale relative to a g-unit: two groups of courses can be distinguished as those with gains (slopes) that do and do not exceed the g-unit.

5. Discussion

Sometimes the most straightforward insights are the easiest to overlook. The Rasch Model (and its family of extensions) provide an attractive analogy to measurement in the physical sciences [28,14]. But this analogy does nothing to guarantee that the default unit of the model, the logit, can be used to meaningfully interpret a contrast between two locations on the scale. The first order value of a graphic such as the Wright Map is that it allows the location of people to be compared with the location of items. However, the purpose of the Rasch Model is usually to facilitate comparisons between people (and between items), and for this purpose it can be valuable to establish two benchmarks on the scale and use these benchmarks to define a more substantively interpretable reference unit. Two strategies were explored to illustrate how this could be accomplished. In the first, benchmarks could be defined with respect to item design factors. In the second, benchmarks could

be defined according to learning targets. Other strategies would be possible, but the common thread is that the strategy be based on a theory connecting the variability in the person attribute of interest to variability in the difficulty of items on the test instrument. In the present instance, benchmarks based on learning targets were used to define a reference unit for growth (the g-unit) as a ratio of the original logit unit in a default Rasch Model calibration. This g-unit was then used to visualize and interpret the magnitude of average learning gains for students who took an assessment with a pre-post design.

Advocates of the Rasch Model as a paradigm for measurement in the human sciences may do a disservice to their position when they promote a “Wright Map” visualization as an analog to the ruler (or to a thermometer) without giving careful thought to both the aesthetics of their design and their choice of reference units. In the measurement of physical attributes, the initial choice of a standard unit has often been made arbitrarily, but once that choice is established, the human mind can and will associate it with real and observable phenomena in the world. These observable phenomena become internalized as meaningful frames of reference. In contrast, when attempting to measure psychological attributes, it can be argued that the choice of a standard unit should not be made arbitrarily, but intentionally. The intention should be to select a unit that can be evaluated against some testable theory of the attribute. In the illustration provided earlier, this was done by checking whether the g-unit distance remained invariant when estimated on the basis of students exposed to a different curricular sequence in their introductory genetics courses. Findings of invariance will indicate constraints on the generalizability of a proposed unit for measurement, at least until the theory explaining variability in the attribute is better understood.

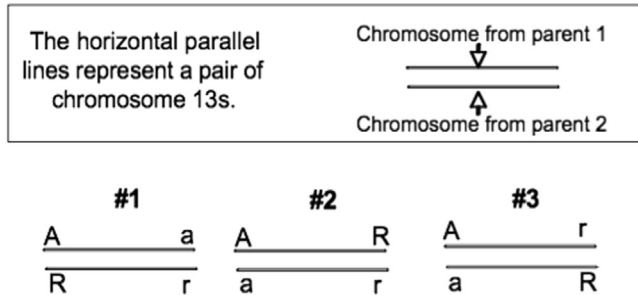
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Appendix

Fig. A-1. GCA Items Used to Establish Reference Unit.

13. A man is a carrier for Wilson's disease (Aa) and Rotor syndrome (Rr). Assume the genes involved in these two disorders are both on chromosome 13 (a non-sex chromosome). Below are possible representations of his genotype (labeled #1, #2, and #3). Which of them could be correct?



- a) #1 only
- b) #2 only
- c) #3 only
- d) #2 and #3 only
- e) #1, #2 and #3

24. Two different genes are located on the same chromosomal pair in rabbits. A particular female rabbit is heterozygous for alleles of both these genes, with the alleles arranged as shown in the diagram to the right. Scientists know that the two genes are on the same chromosome, but do not know their exact position, as indicated by the dashed line.



Suppose this female mates with a male rabbit in which the same chromosome pair looks like this:



How likely is it that this pair of rabbits would have offspring with a chromosome pair that looks like this:



- a) Not likely, because the R and e alleles are not on the same chromosome in either parent.
- b) Very likely, because the random assortment of chromosomes during cell division to make sperm or eggs allows for the mixing of all alleles.
- c) More likely if the two genes are very close together on the chromosome.
- d) More likely if the two genes are not very close together on the chromosome.

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